HW2.2

SCIE 001 Math

Noah Virjee
45515863

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1. (a)

> For this problem, consider the integral:
>
> $$S = \int_0^{2\pi} \sin^k(x)\,\mathrm{d}x$$
>
> If k is an odd positive integer, determine the value of S.

$$S = \int_0^{2\pi} \sin^k(x)\,\mathrm{d}x = \int_0^{2\pi} \sin^{k-1}(x)\sin^1(x)\,\mathrm{d}x$$

All odd numbers subtract 1 are even. And all even numbers are divisible by 2, and hence are representable by some integer times 2.

Since k is odd, $k - 1$ must be even. Therefore, we can let $2p = k - 1$, where p is some positive integer such that $p \geq 0$.

$$\begin{aligned} S &= \int_0^{2\pi} \sin^{2p}(x)\sin^1(x)\,\mathrm{d}x \\ &= \int_0^{2\pi} \left(\sin^2(x)\right)^p \sin^1(x)\,\mathrm{d}x \\ &= \int_0^{2\pi} \left(1 - \cos^2(x)\right)^p \sin^1(x)\,\mathrm{d}x \qquad \text{since } \sin^2(x) + \cos^2(x) = 1 \end{aligned}$$

u-substitution for $\cos(x)$:

$$u = \cos(x) \qquad \mathrm{d}u = -\sin(x)\,\mathrm{d}x$$

$$\begin{aligned} S &= \int_0^{2\pi} -\left(1 - \cos^2(x)\right)^p (-\sin^x)\,\mathrm{d}x \\ &= -\int_{\cos(0)}^{\cos(2\pi)} \left(1 - u^2\right)^p \mathrm{d}u \\ &= -\int_1^1 \left(1 - u^2\right)^p \mathrm{d}u \end{aligned}$$

Any integral from one number to itself is 0. So:

$$S = -\int_1^1 \left(1 - u^2\right)^p \mathrm{d}u = 0$$

Therefore for any odd positive integer k:

$$\boxed{\mathrm{S} = 0}$$

(a) Calculate

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x$$

i. by using a u-substitution.
ii. by using a trig substitution.

Then compare your results.

First a u-substitution:

$$\begin{aligned} u &= \sqrt{a^2-x^2} & u^2 &= a^2-x^2 \\ \mathrm{d}u &= \left(\frac{1}{2}\right)\left(a^2-x^2\right)^{-\frac{1}{2}}(-2x)\,\mathrm{d}x & x^2 &= a^2-u^2 \\ \mathrm{d}u &= \frac{-x}{\sqrt{a^2-x^2}} & x &= \pm\sqrt{a^2-u^2} \end{aligned}$$

$$\begin{aligned} &\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x \\ &= \int \left(\frac{\sqrt{a^2-x^2}}{x}\right)\left(\frac{-x\sqrt{a^2-x^2}}{-x\sqrt{a^2-x^2}}\right)\mathrm{d}x \\ &= \int \left(\frac{\left(\sqrt{a^2-x^2}\right)^2}{-x^2}\right)\left(\frac{-x}{\sqrt{a^2-x^2}}\right)\mathrm{d}x \\ &= \int \frac{u^2}{-\left(\pm\sqrt{a^2-u^2}\right)^2}\,\mathrm{d}u \\ &= -\int \frac{u^2}{a^2-u^2}\,\mathrm{d}u \end{aligned}$$

Now we do a quick polynomial division:

$$\begin{array}{r|rrr} & & & -1 \\ \hline -u^2+a^2 & u^2 & +0u & +0 \\ & -(u^2 & +0u & -a^2) \\ & & & a^2 \end{array}$$

Rewriting:

$$\begin{aligned} &-\int \frac{u^2}{a^2-u^2}\,\mathrm{d}u \\ &= -\int -1 + \frac{a^2}{a^2-u^2}\,\mathrm{d}u \\ &= -\left(-u + \int \frac{a^2}{a^2-u^2}\,\mathrm{d}u\right) \\ &= u - \int \frac{a^2}{a^2-u^2}\,\mathrm{d}u \end{aligned}$$

Followed by some partial fractions:

$$\frac{a^2}{a^2-u^2} = \frac{a^2}{(a+u)(a-u)} = \frac{A}{a+u} + \frac{B}{a-u}$$

$$A(a-u)+B(a+u) = a^2$$

$$Aa - Au + Ba + Bu = a^2$$

$$\begin{aligned} Aa + Ba &= a^2 & -Au + Bu &= 0u \\ A + B &= a & -A + B &= 0 \\ 2B &= a & B &= A \\ B &= \frac{a}{2} & A &= \frac{a}{2} \end{aligned}$$

So we can rewrite:

$$\begin{aligned} & u - \int \frac{a^2}{a^2-u^2}\,\mathrm{d}u \\ &= u - \int \frac{\frac{a}{2}}{a+u} + \frac{\frac{a}{2}}{a-u}\,\mathrm{d}u \\ &= u - \left(\frac{a}{2}\ln|a+u| - \frac{a}{2}\ln|a-u|\right) + C \\ &= u - \frac{a}{2}\ln|a+u| + \frac{a}{2}\ln|a-u|) + C \end{aligned}$$

Since $u = \sqrt{a^2-x^2}$, finally:

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = \sqrt{a^2-x^2} - \frac{a}{2}\ln\left|a+\sqrt{a^2-x^2}\right| + \frac{a}{2}\ln\left|a-\sqrt{a^2-x^2}\right| + C$$

Now with trigonometric substitution:

$$x = a\sin\theta \qquad \mathrm{d}x = a\cos\theta\,\mathrm{d}\theta \qquad \theta = \arcsin\left(\frac{x}{a}\right)$$

$$\begin{aligned} & \int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x \\ &= \int \frac{1}{a\sin\theta}\sqrt{a^2-(a\sin\theta)^2}a\cos\theta\,\mathrm{d}\theta \\ &= \int \frac{\cos\theta}{\sin\theta}\sqrt{a^2-(a^2\sin^2\theta)}\,\mathrm{d}\theta \\ &= \int \frac{\cos\theta}{\sin\theta}\sqrt{a^2-(a^2-a^2\cos^2\theta)}\,\mathrm{d}\theta \qquad \text{using } \sin^2\theta = 1-\cos^2\theta \\ &= \int \frac{a\cos^2\theta}{\sin\theta}\,\mathrm{d}\theta \\ &= a\int \frac{1-\sin^2\theta}{\sin\theta}\,\mathrm{d}\theta \qquad \text{using } \cos^2\theta = 1-\sin^2\theta \\ &= a\int \frac{1}{\sin\theta} - \sin\theta\,\mathrm{d}\theta \\ &= a\cos\theta + a\int \frac{1}{\sin\theta}\,\mathrm{d}\theta \end{aligned}$$

The integral of $\frac{1}{\sin\theta}$ can be done separately:

$$\begin{aligned}
\int \frac{1}{\sin\theta}\,\mathrm{d}\theta &= \int \frac{\sin\theta}{\sin^2\theta}\,\mathrm{d}\theta \\
&= \int \frac{\sin\theta}{1-\cos^2\theta}\,\mathrm{d}\theta \qquad \text{using } \sin^2\theta = 1-\cos^2\theta \\
&= \int -\frac{-\sin\theta}{1-\cos^2\theta}\,\mathrm{d}\theta \qquad u=\cos\theta \qquad \mathrm{d}u = -\sin\theta \\
&= -\int \frac{1}{1-u^2}\,\mathrm{d}u
\end{aligned}$$

This becomes a partial fractions:

$$\frac{1}{1-u^2} = \frac{1}{(1+u)(1-u)} = \frac{A}{1+u} + \frac{B}{1-u} = 1$$
$$A(1-u) + B(1+u) = 1$$
$$A - Au + B + Bu = 1$$

$$\begin{aligned}
A+B &= 1 \qquad & -Au + Bu &= 0u \\
A+A &= 1 \qquad & A &= B \\
A &= \frac{1}{2} \qquad & B &= \frac{1}{2}
\end{aligned}$$

So we can rewrite:

$$\begin{aligned}
\int \frac{1}{\sin\theta}\,\mathrm{d}\theta &= -\int \frac{\frac{1}{2}}{1+u} + \frac{\frac{1}{2}}{1-u}\,\mathrm{d}\theta \\
&= -\left(\frac{1}{2}\ln|1+u| - \frac{1}{2}\ln|1-u|\right) + C \\
&= -\frac{1}{2}\ln|1+\cos(\theta)| + \frac{1}{2}\ln|1-\cos(\theta)| + C
\end{aligned}$$

Returning to our original integral:

$$\begin{aligned}
&\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x \\
&= a\cos\theta + a\int \frac{1}{\sin\theta}\,\mathrm{d}\theta \\
&= a\cos\theta + a\left(-\frac{1}{2}\ln|1+\cos(\theta)| + \frac{1}{2}\ln|1-\cos(\theta)| + C\right)
\end{aligned}$$

Since $\theta = \arcsin\left(\frac{x}{a}\right)$, finally:

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = a\cos\left(\arcsin\left(\frac{x}{a}\right)\right) + a\left(-\frac{1}{2}\ln\left|1+\cos\left(\left(\arcsin\left(\frac{x}{a}\right)\right)\right)\right| + \frac{1}{2}\ln\left|1-\cos\left(\left(\arcsin\left(\frac{x}{a}\right)\right)\right)\right| + C\right)$$

However, this can be simplified. As we know:

$$\cos^2\theta + \sin^2\theta = 1$$
$$\cos\theta = \pm\sqrt{1-\sin^2\theta}$$

"look into this"

Since $\theta = \arcsin\left(\frac{x}{a}\right)$:

$$\cos\left(\arcsin\left(\frac{x}{a}\right)\right) = \pm\sqrt{1-\sin^2\left(\arcsin\left(\frac{x}{a}\right)\right)}$$

$$= \pm\sqrt{1-\left(\frac{x}{a}\right)^2}$$

$$\cos\left(\arcsin\left(\frac{x}{a}\right)\right) = \pm\sqrt{1-\frac{x^2}{a^2}}$$

Plugging this into our expression:

Look into the loss of the plus or minus here!

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = a\cos\left(\arcsin\left(\frac{x}{a}\right)\right) + a\left(-\frac{1}{2}\ln\left|1+\cos\left(\left(\arcsin\left(\frac{x}{a}\right)\right)\right)\right| + \frac{1}{2}\ln\left|1-\cos\left(\left(\arcsin\left(\frac{x}{a}\right)\right)\right)\right| + C\right)$$

$$= a\sqrt{1-\frac{x^2}{a^2}} + a\left(-\frac{1}{2}\ln\left|1+\sqrt{1-\frac{x^2}{a^2}}\right| + \frac{1}{2}\ln\left|1-\sqrt{1-\frac{x^2}{a^2}}\right|\right) + C$$

$$= \sqrt{a^2\left(1-\frac{x^2}{a^2}\right)} + a\left(-\frac{1}{2}\ln\left|1+\sqrt{1-\frac{x^2}{a^2}}\right| + \frac{1}{2}\ln\left|1-\sqrt{1-\frac{x^2}{a^2}}\right|\right) + C$$

Now we can do some rearranging and logarithm shenanigans:

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = \sqrt{a^2-x^2} + \frac{a}{2}\left(-\ln\left|1+\sqrt{1-\frac{x^2}{a^2}}\right| + \ln\left|1-\sqrt{1-\frac{x^2}{a^2}}\right|\right) + C$$

$$= \sqrt{a^2-x^2} + \frac{a}{2}\left(-\ln\left|1+\sqrt{1-\frac{x^2}{a^2}}\right| + \ln\left|1-\sqrt{1-\frac{x^2}{a^2}}\right| + \ln|a| - \ln|a|\right) + C$$

$$= \sqrt{a^2-x^2} + \frac{a}{2}\left(-\left(\ln\left|1+\sqrt{1-\frac{x^2}{a^2}}\right| + \ln|a|\right) + \left(\ln\left|1-\sqrt{1-\frac{x^2}{a^2}}\right| + \ln|a|\right)\right) + C$$

$$= \sqrt{a^2-x^2} + \frac{a}{2}\left(-\ln\left|a\left(1+\sqrt{1-\frac{x^2}{a^2}}\right)\right| + \ln\left|a\left(1-\sqrt{1-\frac{x^2}{a^2}}\right)\right|\right) + C$$

$$= \sqrt{a^2-x^2} + \frac{a}{2}\left(-\ln\left|a+\sqrt{a^2-x^2}\right| + \ln\left|a-\sqrt{a^2-x^2}\right|\right) + C$$

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = \sqrt{a^2-x^2} - \frac{a}{2}\ln\left|a+\sqrt{a^2-x^2}\right| + \frac{a}{2}\ln\left|a-\sqrt{a^2-x^2}\right| + C$$

Comparing the two results:

Result 1:

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = \sqrt{a^2-x^2} - \frac{a}{2}\ln\left|a+\sqrt{a^2-x^2}\right| + \frac{a}{2}\ln\left|a-\sqrt{a^2-x^2}\right| + C$$

Result 2:

$$\int \frac{1}{x}\sqrt{a^2-x^2}\,\mathrm{d}x = \sqrt{a^2-x^2} - \frac{a}{2}\ln\left|a+\sqrt{a^2-x^2}\right| + \frac{a}{2}\ln\left|a-\sqrt{a^2-x^2}\right| + C$$

Result 1 is the same as result 2, therefore the method that you take to solve this integral does not impact the result.

2. In tutorial, we looked at metapopulations and the Levins model derived by Richard Levins in 1969 which describes metapopulations as a means for studying spatially structured populations. In particular, the model tracks the proportion of patches that are occupied by the population, but does not track the density of the population nor which specific patches are inhabited. These subpopulations are modelled through the Levins model

$$\frac{\mathrm{d}p}{\mathrm{d}t} = cp(1-p) - mp$$

where $p = p(t)$ is the fraction of patches occupied, $m > 0$ is the mortality of a subpopulation, and $c > 0$ is the colonization of a vacant subpopulation. Recall that vacant patches can be colonized at a rate proportional to the fraction of occupied patches as $1 - p$.

(a) Set $m = 1$ and $c = 3$.

i. Algebraically solve, using an appropriate integration technique, the Levins model and determine the long term behaviour (i.e., equilibria). Be sure to include all steps in your solution.

Setting $m = 1$ and $c = 3$:

$$\begin{aligned}\frac{\mathrm{d}p}{\mathrm{d}t} &= 3p(1-p) - (1)p \\ &= 3p - 3p^2 - p \\ \frac{\mathrm{d}p}{\mathrm{d}t} &= -3p^2 + 2p\end{aligned}$$

Integrating:

$$\begin{aligned}\int \frac{1}{-3p^2+2p}\,\mathrm{d}p &= \int 1\,\mathrm{d}t \\ t &= \int \frac{1}{p(-3p+2)}\,\mathrm{d}p\end{aligned}$$

Using partial fractions:

$$\frac{1}{p(-3p+2)} = \frac{A}{p} + \frac{B}{-3p+2}$$

$$A(-3p+2) + Bp = 1$$

$$-3pA + Bp = 0 \qquad 2A = 1$$

$$-3A + B = 0 \qquad A = \frac{1}{2}$$

$$B = 3A \qquad B = \frac{3}{2}$$

So our integral becomes:

$$t = \int \frac{\frac{1}{2}}{p} + \frac{\frac{3}{2}}{-3p+2} \, \mathrm{d}p$$

$$t = \int \frac{\frac{1}{2}}{p} \, \mathrm{d}p + \int \frac{-3(-\frac{1}{2})}{-3p+2} \, \mathrm{d}p \qquad u = -3p + 2 \qquad \mathrm{d}u = -3 \, \mathrm{d}p$$

$$t = \left(\frac{1}{2}\right) \ln|p| + \int \frac{-\frac{1}{2}}{u} \, \mathrm{d}u$$

$$t = \frac{1}{2} \ln|p| - \frac{1}{2} \ln|u| + C$$

$$t = \frac{1}{2} \ln|p| - \frac{1}{2} \ln|-3p+2| + C$$

$$t = \frac{1}{2} (\ln|p| - \ln|-3p+2|) + C$$

$$t = \frac{1}{2} \ln\left|\frac{p}{-3p+2}\right| + C$$

$$t = \frac{1}{2} \ln\left|\frac{1}{-3 + \frac{2}{p}}\right| + C$$

Solving for p:

$$t = \frac{1}{2} \ln\left|\frac{1}{-3 + \frac{2}{p}}\right| + C$$

$$e^{2(t-C)} = \left|\frac{1}{-3 + \frac{2}{p}}\right|$$

We are attempting to find an expression for p given t so we can remove the absolute value bars and replace them with a plus or minus:

$$e^{2(t-C)} = \frac{\pm 1}{-3 + \frac{2}{p}}$$

$$-3 + \frac{2}{p} = \frac{\pm 1}{e^{2(t-C)}}$$

$$\frac{2}{p} = \frac{1}{\pm e^{2(t-C)}} + 3$$

$$p = \frac{2}{\pm e^{-2(t-C)} + 3}$$

Looking at the equilibria (where $\frac{\mathrm{d}p}{\mathrm{d}t} = 0$):

$$\frac{\mathrm{d}p}{\mathrm{d}t} = -3p^2 + 2p = 0$$

$$p(-3p+2) = 0$$

$$p = 0 \qquad \text{or} \qquad p = \frac{2}{3}$$

When $0 < p < \frac{2}{3}$:

$$\begin{aligned}
p &< \frac{2}{3} \\
-3p &> -2 && \text{the comparison flips since } -3 < 0 \\
-3p + 2 &> 0 \\
p &> 0 \\
p(-3p + 2) &> 0 \\
\therefore \frac{\mathrm{d}p}{\mathrm{d}t} &> 0
\end{aligned}$$

When $p > \frac{2}{3}$:

$$\begin{aligned}
p &> \frac{2}{3} \\
-3p &< -2 && \text{the comparison flips since } -3 < 0 \\
-3p + 2 &< 0 \\
p &> 0 \\
p(-3p + 2) &< 0 \\
\therefore \frac{\mathrm{d}p}{\mathrm{d}t} &< 0
\end{aligned}$$

So when the population is between 0 and $\frac{2}{3}$ it will increase over time, and when the population is above $\frac{2}{3}$ it will decrease over time. I.e. $p = 0$ is an unstable equilibrium and p = $\frac{2}{3}$ is a stable equilibrium.

ii.
> Interpret your answer in the context of the biological setting of the model: i.e., what are the biological significances of the results you have obtained.

When the mortality of a subpopulation m is 1 and the colonization rate of a vacant subpopulation c is 3, the fraction of patches occupied p will approach $\frac{2}{3} \approx \%66.67$. This holds for all initial conditions of p (for all starting fractions of patches) except for when there is no initial inhabited patches $p = 0$.

Our expression for $p(t)$ also holds some significance:

$$p = \frac{2}{\pm e^{-2(t-C)} + 3}$$

If we consider only the case where:

$$p = \frac{2}{e^{-2(t-C)} + 3}$$

This tracks a sigmoid curve as a small fraction of inhabited patches increases the number of patches slowly at first, limited by the number of patches, then quickly as the number of patches grows, then slowly again as the mortality catches up and it reaches an equilibrium.

(b) If we set $m = 0$ in the Levins model, then we are left with the Logistic Growth model on the proportion of patches. In this case, from our understanding of the Logistic Growth model, we know that all patches will be filled in the long run ($p \to 1$). This observation, along with others related to metapopulations, leads to the need to have $m > 0$. With some manipulation of the right-hand side of the Levins model, we can rewrite it as a Logistic Growth model on the proportion of patches, and use our knowledge of the Logistic Growth model in order to understand the Levins model.

i. Manipulate the right-hand side of the Levins model to put it into the form of the Logistic growth model

$$\frac{\mathrm{d}p}{\mathrm{d}t} = rp\left(1 - \frac{p}{K}\right)$$

and identify the new parameters r and K in terms of m and c.

$$\begin{aligned}
\frac{\mathrm{d}p}{\mathrm{d}t} &= cp(1-p) - mp \\
&= cp - cp^2 - mp \\
&= cp - mp - cp^2 \\
&= p(c-m) - cp^2 \\
&= p(c-m)\left(1 - \frac{cp^2}{p(c-m)}\right) \\
&= p(c-m)\left(1 - \frac{cp}{c-m}\right) \\
&= p(c-m)\left(1 - \left(\frac{cp}{c-m}\right)\right) \\
&= (c-m)p\left(1 - \left(\frac{p}{1-\frac{m}{c}}\right)\right)
\end{aligned}$$

Comparing this to the logistic growth equation:

$$\begin{aligned}
\frac{\mathrm{d}p}{\mathrm{d}t} &= rp\left(1 - \frac{p}{K}\right) \\
\frac{\mathrm{d}p}{\mathrm{d}t} &= (c-m)p\left(1 - \left(\frac{p}{1-\frac{m}{c}}\right)\right)
\end{aligned}$$

This gives:

$$\begin{aligned}
r &= c - m \\
K &= 1 - \frac{m}{c}
\end{aligned}$$

ii. Interpret the effects of the parameters m and c as they relate to the Logistic Growth model.

We know that in the Logistic Growth model K represents to the carrying capacity of the ecosystem, i.e. the maximum population size that the system can sustain over time.

The Logistic Growth model's population size p is analogous to the fraction of inhabited patches p n the Levins Model, and as shown, K is analogous to $1 - \frac{m}{c}$. Therefore, in relation to the Logistic Growth model $1 - \frac{m}{c}$ represents the maximum fraction of inhabited patches that the system can sustain over time, i.e. the fraction of inhabited populations over time will approach one minus the mortality rate over the colonization rate. This makes sense - the fraction of inhabited patches should be dependant on the rate that fractions are removed (mortality) divided by the rate that fractions are gained (colonization), and subtracted from the maximum value of p, therefore $K = 1 - \frac{m}{c}$.

Additionally, in the Logistic Growth model r relates to the growth rate, i.e. how fast the population increases or decreases to its carrying capacity. As the population size decreases $\frac{\mathrm{d}p}{\mathrm{d}t}$ approaches pr.

Relating this to the Levins Model, we showed r is analogous to $c - m$. This also makes sense, the growth rate of the fraction of inhabited patches should be the dependant on the rate that fractions are gained (colonization) subtract the rate that fractions are removed (mortality).

3. The Gamma function $\Gamma(x)$ is a continuous function defined by the improper integral

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} \,\mathrm{d}t$$

(a) Compute $\Gamma(1)$.

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty t^{1-1} e^{-t} \,\mathrm{d}t \\
&= \int_0^\infty (1) e^{-t} \,\mathrm{d}t \\
&= \lim_{T\to\infty} \int_0^T e^{-t} \,\mathrm{d}t \\
&= \lim_{T\to\infty} -e^{-t}\Big|_0^T \\
&= \lim_{T\to\infty} -e^{-T} - (-e^{-0}) \\
&= \lim_{T\to\infty} (-e^{-T}) + 1 \\
&= 0 + 1 \\
\therefore \Gamma(1) &= 1
\end{aligned}$$

(b) Use integration by parts to show that $\Gamma(x+1) = x\Gamma(x)$ for $x > 0$.

$$\begin{aligned}
\Gamma(x+1) &= \int_0^\infty t^{(x+1)-1} e^{-t} \,\mathrm{d}t \\
&= \lim_{T\to\infty} \int_0^T t^x e^{-t} \,\mathrm{d}t
\end{aligned}$$

Using integration by parts:

$$u = t^x \qquad \mathrm{d}v = e^{-t}\,\mathrm{d}t$$
$$\mathrm{d}u = xt^{x-1}\,\mathrm{d}t \qquad v = -e^{-t}$$

$$\begin{aligned}\Gamma(x+1) &= \lim_{T\to\infty}\left(-t^x e^{-t}\big|_0^T - \int_0^T -e^{-t}xt^{x-1}\,\mathrm{d}t\right)\\ &= \lim_{T\to\infty}\left(-T^x e^{-T} - \left(-(0)^x e^{-0}\right)\right) + \lim_{T\to\infty}\left(x\int_0^T e^{-t}t^{x-1}\,\mathrm{d}t\right)\\ &= \lim_{T\to\infty}\left(-\frac{T^x}{e^T}\right) + x\lim_{T\to\infty}\left(\int_0^T e^{-t}t^{x-1}\,\mathrm{d}t\right)\end{aligned}$$

If $x > 1$, we can apply L'Hopital's rule and differentiate both sides of the quotient. And noting that $\lim_{T\to\infty}\int_0^T f(x)\,\mathrm{d}x = \int_0^\infty f(x)\,\mathrm{d}x$, we can rewrite:

$$\Gamma(x+1) = \lim_{T\to\infty}\left(-\frac{xT^{x-1}}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t$$

If $x > 2$ we can do L'Hopital's rule again to give us:

$$\Gamma(x+1) = \lim_{T\to\infty}\left(-\frac{x(x-1)T^{x-2}}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t$$

For any x, do L'Hopital's rule x times, giving you:

$$\Gamma(x+1) = \lim_{T\to\infty}\left(-\frac{x(x-1)(x-2)(x-3)(...)T^{x-x}}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t$$

Noting that $x(x-1)(x-2)(x-3)(...)$ exactly x times is equal to $x!$, and $x!$ is finite:

$$\begin{aligned}\Gamma(x+1) &= \lim_{T\to\infty}\left(-\frac{x!T^0}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t\\ &= \lim_{T\to\infty}\left(-\frac{x!}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t\\ &= \lim_{T\to\infty}\left(-\frac{x!}{e^T}\right) + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t\\ &= 0 + x\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t\end{aligned}$$

Therefore:

$$\Gamma(x+1) = x\left(\int_0^\infty e^{-t}t^{x-1}\,\mathrm{d}t\right)$$

Since we know that:

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$$

We have shown that:

$$\Gamma(x+1) = x\Gamma(x)$$

(c) Using the fact that $\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$, compute $\Gamma\left(\frac{1}{2}\right)$. Hint: Make a substitution.

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right) &= \int_0^\infty t^{\left(\frac{1}{2}\right)-1} e^{-t}\,\mathrm{d}t \\ &= \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t \qquad u = \sqrt{t} \qquad \mathrm{d}u = \left(\frac{1}{2}\right)t^{-\frac{1}{2}}\,\mathrm{d}t \qquad t = u^2 \\ &= \int_0^\infty (2)e^{-u^2}\,\mathrm{d}u \\ &= 2\int_0^\infty e^{-u^2}\,\mathrm{d}u\end{aligned}$$

Given $\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$:

$$\Gamma\left(\frac{1}{2}\right) = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2\left(\frac{\sqrt{\pi}}{2}\right)$$

$$\therefore \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

(d) Find $\Gamma\left(\frac{3}{2}\right)$.

In question b we showed that $\Gamma(x+1) = x\Gamma(x)$ for $x > 0$, and in question c we found that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. So we can express our problem to take advantage of this:

$$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)\sqrt{\pi}$$

Therefore:

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

Remark: Using the recurrence relation in (b), it can be proved that $\Gamma(n+1) = n!$ when n is a positive integer. (Recall that the factorial is defined as $n! = n \cdot (n-1) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$, but this formula is only valid when n is a natural number.)

In view of this property,$\Gamma(x+1)$ is often written as $x!$ and regarded as an extension of the factorial function to real numbers. Some scientific calculators with the factorial function $n!$ built in actually calculate the gamma function rather than the simpler formula of products of integers. Check whether your calculator does this by asking it for 0.5!. If you get an error message, it is not using the gamma function. If you get a numerical value, compare it with your answer in (c).